Metropolitan University
Department of Computer Science and Engineering
Summer Semester Final Examination – 2023
Program: B.Sc. in CSE; Batch: 57^{th} (All Section)
Course: CSE 215: Communication Engineering

Time: 2 Hours **Marks: 40**

Answer any **four** questions. All parts of each question must be answered sequentially. Figures in the right margin indicate full marks.

1. (a) Figure out GSM Architecture. 3
   (b) What is cell? Discuss multiple Access technique with figure. 1+3
   (c) Write the radio parameter and characteristics of GSM-900. 3

2. (a) Draw General Packet Radio Service (GPRS). 3
   (b) Write the functionalities of BTS and BSS. 4
   (c) Discuss working procedure of SIM with figure. 3

3. (a) Describe call termination steps of GSM with figure. 4
   (b) Write the functionalities of MSC and HLR. 4
   (c) Write short note on CAMEL and HSCSD. 2

4. (a) Write the steps of mobile call origination with figure. 5
   (b) Write the steps of speech encoding with figure. 5

5. (a) Write the steps of mobile inter MSC Handoff mechanism with figure. 5
   (b) Discuss logical channel with figure of GSM-900. 5

6. (a) Write about (i) New supplementary services (ii) New features and Capabilities of GSM. 2
   (b) Discuss protocol model with figure of GSM-900. 5
   (c) Discuss Location Update System with figure of GSM-900. 3

Metropolitan University, Sylhet

Department of Computer Science and Engineering

Summer Semester Final Examination – 2023

Program: B.Sc. in CSE; **Batch: 57(A, B, C, D, E, F&G)**

Course: GED-213: **Principles of Economics & Entrepreneurship Development**

Time: 2 Hours ***Full Marks: 40***

Answer any **four** questions. All parts of each question must be answered sequentially.

Figures in the right margin indicate full marks.

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1. a) Identify the primary elements of a new-venture team. 1.5
   b) Differentiate between inside directors and outside directors. 2
   c) What is the difference between heterogeneous and homogeneous founding team? 2.5
   d) Discuss the purpose of forming an advisory board. 4

2. a) Identify and describe techniques entrepreneurs use to generate ideas. 2.5
   b) Explain the three types of start-up firms. 2.5
   c) Describe the five common myths regarding entrepreneurships. 5

3. a) In developing a successful business idea, summarize the role of feasibility analysis. 3
   b) Broadly describe the four areas of feasibility analysis of a business idea. 7

4. a) What are the reasons for preparing a business plan for new ventures? 2
   b) What are the types of business plan? 2
   b) Demonstrate the outline of a business plan. 6

5. a) Fill in blanks in table below

		Year 1(2000) (Base year)		Year 2 (2003)		Year 3 (2006)	
Market Basket of Items	Units	$ per unit	Total Cost	$ per unit	Total Cost	$ per unit	Total Cost
Cheese	2 lbs	$1.75	$3.50	$1.50	$ ---	$1.50	$ ---
Blue Jeans	2 pairs	$12.00	$ ---	$15.50	$ ---	$20.00	$ ---
Gasoline	10 gallons	$1.25	$ ---	$1.60	$ ---	$2.70	$ ---
Total Cost of Basket		-------	------	------	------	-------	-------

Using the above table Calculate:

a) Calculate the index for each year. 5

b) Calculate the inflation rate on a percentage basis for the following: i) Year 1 to Year 2 ii) Year 1 to Year 3 and iii) Year 2 to Year 3. 5

6. a) You are given the following table:

Year	Nominal GDP	GDP Deflator
2022	$5,000	125
2023	$6,600	150

4000
4400

By using above table calculate: Real GDP in year 2022 and Year 2023. 6

b) Explain the circular flow of income in two-sector economy with diagram. 4

Metropolitan University

Department of Computer Science and Engineering

Summer Semester Final Examination – 2023

Program: B.Sc. in CSE; **Batch: 57**

Course: CSE 221: Object Oriented Programming

Time: 2 Hours **Marks: 40**

Answer any **four** questions. All parts of each question must be answered sequentially. Figures in the right margin indicate full marks.

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1. a) Explain the four main principles of OOP and how they are implemented in Java. 5

   b) Define and compare between the **super** and **this** keywords. 3

   c) Explain and demonstrate the significance of the **static** keyword in Java. 2

2. a) Explain how OOP promotes code reusability compared to structured programming. Mention the roles of inheritance and polymorphism in your answer. 4

   b) Discuss how access modifiers contribute to encapsulation and inheritance in Java. 4

   c) What is multithreading, and how does it differ from single-threading? 2

3. a) What are runtime errors? Illustrate with a code snippet how we can handle them. 4

   b) Compare between the **final** and **abstract** keywords in Java. 3

   c) Discuss the difference between compile-time polymorphism and runtime polymorphism. 3

4. a) Define interface and abstract class in OOP. Compare abstract classes and interfaces. 4

   b) Discuss the concept of garbage collection in Java. Highlight its significance in memory management. 3

   c) Explain the three ways in which the **final** keyword can be used. 3

5. Solve the following problem with all the functionality mentioned below 10

- Create an abstract class called **Shape** with **abstract** methods **calculateArea()** and **calculatePerimeter()**.
- Implement a subclass named **Rectangle** that inherits from the Shape class, which takes two parameters, **height** and **width**.
- Implement the **calculateArea()** and **calculatePerimeter()** methods for Rectangle class.
- Write a **Test** class to create an object of **Rectangle** class.
- Call the two methods, and **Print** the output.

6. Solve the following problem with all the functionality mentioned below 10

- Create a **Student** class with the attribute name.
- Add two encapsulated attributes, **age** and **CGPA**, to the class.
- Implement proper methods to **set** and **get** the encapsulated variables.
- Implement a special condition where the age cannot be set if it is below 18.
- Write a **Test** class to instantiate a **Student** object and print all the information.

Metropolitan University, Sylhet

Department of Computer Science and Engineering

Summer Semester Final Examination 2023

Program: BSc in CSE: **Batch 57** ([redacted]) all batch

Course: MAT 135, Matrix, Complex Variable and Fourier analysis

Time: 2 Hours **Marks: 40**

All parts of each question must be answered sequentially. Figures in the right margin indicate full marks

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(Answer any four Questions)

1. (a) Define Periodic matrix and Hermitian matrix. 4

   (b) Prove that, every square matrix may be written as the sum of Hermitian and Skew Hermitian matrices. 6

2. (a) Define Limit point and Analytic function. 3

   (b) Find the necessary conditions that, the complex function f(z) to be Analytic. 7

3. (a) Find the value of $\int_{PQ} z\,dz$, where z is a complex number. 5

   (b) Find the value of the integral $\int_{0}^{1+i}(x-y+ix^2)dz$ along the straight line from z = 0 to z = 1+i. 5

4. (a) State and prove (elementary proof) the Cauchy's theorem for an analytic function f(z). 5

   (b) If f(z) is an analytic in a region by two simple closed curve C and C_1 (C_1 lies inside C) and on this curve, then $\oint_C f(z)dz = \oint_{C_1} f(z)dz$, where C and C_1 are both traversed in the positive sense relative to their interiors. 5

5. (a) Write down the Fourier series for the interval $(-\pi,\pi)$ with period 2π. 2

   (b) If f(z) is an even function , then show that (i) $a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)dx$, 8

   $(ii)\, a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos nx\,dx.$, where a_n is Fourier's coefficient.

6. Expand in Fourier sine and cosine series of the function $f(x) = x + x^2$ in the interval $-\pi < x < \pi$. Also show that, $\frac{\pi^2}{6} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \ldots\ldots\ldots\ldots$ 10

Metropolitan University

Department of Computer Science and Engineering

Summer Semester Final Examination - 2023

Program: B.Sc. in CSE (2^{nd} Year 1^{st} Semester); **Batch: 57**

Course: CSE 231: Algorithm Design and Analysis

Time: 2 Hours

Marks: 40

Answer any **four** sets of questions. All parts of each question must be answered sequentially. Figures in the right margin indicate full marks.

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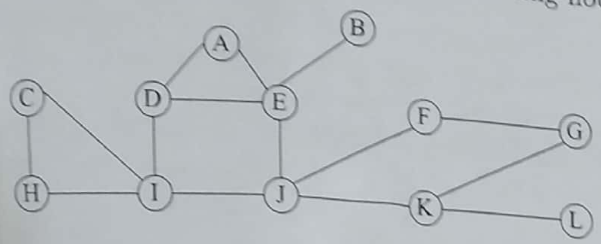
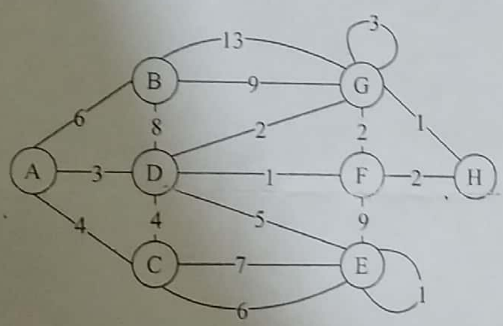
1. Traverse the following graph in BFS and DFS order. Show the implementation process using stack or queue. Assume that A is the starting node. (10)



2. (a) Calculate the first 4 possible solutions of 5 Queen problem(5×5) using Backtracking method. Draw the State Space Tree. (7)

   (b) Consider an edge-connected city graph. The nodes colored red indicate COVID-affected cities, and white-colored nodes indicate healthy cities. You are asked to find out the time the COVID virus will take to affect all the non-affected cities if it takes one unit of time to travel from one city to another. Which method would you choose among BFS and DFS? Justify your answer. (3)

3. You are given such a graph which can provide two or more MSTs. Your task is to find two different Minimum Spanning Trees from the following graph. One MST using Prim's algorithm (Starting Node A) and another MST using Kruskal's algorithm. (10)



4. (a) Apply a 'Single-source Shortest Path' algorithm to the graph provided in question 3 and determine the shortest path from node B to E (Ignore the self-loops in nodes G and E). (5)

   (b) Suppose you are given the following denominations of coin: 7, 2, 4, 3. Apply Dynamic Programming approach to make 15 using maximum number of coins. (5)

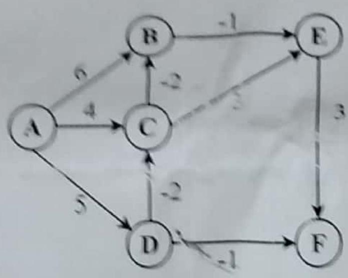
5. (a) Prove that Dijkstra doesn't work when a graph contains negative weighted edge. (3)

(b) Consider the following graph where A is the source node. (5+1+1)

i) Find the shortest path from the source node to all other nodes.

ii) Find the distance and path from A to F.

iii) Is there any negative weight cycle? Justify your answer.



6. (a) Is Backtracking better than Brute Force? Justify your answer. (2)

(b) Imagine designing an automated tour planning system for a personalized adventure tour agency. Each tour visits multiple destinations to maximize customer satisfaction. However, not all destination combinations are possible due to budget and time constraints. Your task is to generate all possible routes within the given constraints, considering different combinations of destinations for each tour. What technique will you follow to complete this task? Justify your answer. (2)

(c) Apply an 'All Pair Shortest Path' algorithm for the following graph (represented as adjacency matrix). Calculate the shortest paths between all pairs. (6)

$$
\begin{array}{c c}
 & \begin{array}{cccc} A & B & C & D \end{array} \\
\begin{array}{c} A \\ B \\ C \\ D \end{array} &
\begin{bmatrix} 0 & 3 & 8 & 5 \\ 11 & 0 & 5 & 6 \\ 12 & 15 & 0 & 7 \\ 5 & 8 & 13 & 0 \end{bmatrix}
\end{array}
$$

Metropolitan University, Sylhet

Department of Computer Science and Engineering
Summer Term Final Examination – 2023
Program: B.Sc. in CSE **Batch:** 57 (All section)
Course: GED 201: Bangladesh Studies

Time: 2 Hours **Marks: 40**

Answer any **four** questions. All parts of each question must be answered sequentially. Figures in the right margin indicate full marks.

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1. (a) You are a policy maker of your hometown, your mission is to involve your community helping you to achieve the UN "Sustainable Development Goals 2030" there, design a campaign and implementation Strategy. 6

   (b) How do you manage a disaster according to the United Nations Office of Disaster Management cycle framework? 4

2. (a) Is GDP and GNP an accurate measurement tool for a country's economic development? Showcase your argument. 5

   (b) Differentiate between industry and sector with an example of your choice. 5

3. (a) Define and distinguish equity and equality, write in your own words- requirements for women empowerment in the contemporary time? 6

   (b) Is women's representation in politics satisfactory? Justify in the context of Bangladesh. 4

4. (a) Write a short note of the opportunities Bangladesh has in the Blue economy. 5

   (b) How do you think Bangladesh can produce clean and renewable energy in the context of the Blue Economy? 4

5. (a) Define demographic dividend and what opportunity Bangladesh has with its one? 5

   (b) What are your recommendations to Bangladeshi policy makers for making Bangladesh a IT tech skill based nation? 5

6. (a) Discuss the pillars of Smart Bangladesh. 5

   (b) How do you want to see Bangladesh in 2041? 5